

# GroveLEDBarPWM — Full API Reference

## Version 2.3.0

Seeed Grove LED Bar V2.1 / MY9221 • Arduino-compatible boards including ESP32-C3

## 1. Overview

GroveLEDBarPWM provides PWM control of ten LEDs on the Grove LED Bar V2.1, including individual brightness, graph levels, global brightness, flashing, effects, transitions and non-blocking updates.

## 2. Installation

Include , create the bar object with DATA and CLOCK GPIO pins, then call begin() from setup().

```
GroveLEDBarPWM bar(7, 6);
```

## 3. Graph Level

setLevel(level) sets the logical graph level from 0 to 10. clear() turns all LEDs off. show() transmits the current frame.

## 4. Individual LED Brightness

setBrightness(index, brightness) sets PWM 0–255. setBrightnessPercent(index, percent) sets 0–100%. getBrightness() and getBrightnessPercent() read an LED's stored brightness.

## 5. setLED() — NEW IN V2.3.0

Sets one logical LED to a percentage brightness. Only the selected LED is changed; other LEDs retain their values. Values above 100% are limited to 100%. Setting an LED also stops a flash on that LED.

```
bar.setLED(3, 75); bar.setLED(3, 0); bar.setLED(3, 100);
```

## 6. Global Brightness

setGlobalBrightness(percent) controls overall output from 0–100% without changing underlying LED brightness. getGlobalBrightness() reads it.

## 7. Flashing LED

flashLED(index) starts a non-blocking flash. The extended form accepts timing and peak brightness. isFlashing() reports flash activity.

```
bar.flashLED(0); bar.flashLED(0, 500, 10); bar.flashLED(0, 500, 30); bar.flashLED(0, 150, 100);
```

## Flash parameters

The milliseconds parameter controls flash timing independently of peak brightness. flashBrightness controls the peak from 0–100%.

## 8. Graduated Brightness

setGraduated(), getGraduated(), setGraduationMin() and getGraduationMin() control graduated graph brightness.

## 9. Green-to-Red Direction

setGreenToRed() controls logical direction; getGreenToRed() reads it.

## 10. Smooth Transitions

setTransition(), setTransitionSpeed(), setTransitionTime() and isTransitioning() control and report smooth transitions.

## 11. Effects

Available effects include EFFECT\_MOVING\_DOT, EFFECT\_KNIGHT\_RIDER and EFFECT\_PULSE. Use startEffect(), stopEffect(), setEffectSpeed() and isEffectRunning().

## 12. Pulse/Breathe

setPulseRange(minimumPercent, maximumPercent) sets the Pulse/Breathe limits.

## 13. update()

Call bar.update() repeatedly from loop() to service transitions, flashes and effects. In the ESP32-C3 application configuration used with this library, manual updates are preferred.

```
void loop() {  bar.update();  // Other application code }
```

## 14. Pin Configuration

setPins(dataPin, clockPin) changes the DATA/CLOCK pins. getDataPin() and getClockPin() read them.

## 15. API Summary

| Function                                                 | Purpose                                        |
|----------------------------------------------------------|------------------------------------------------|
| begin()                                                  | Initialise the LED bar                         |
| setPins()                                                | Change DATA/CLOCK pins                         |
| getDataPin()/getClockPin()                               | Read configured pins                           |
| setLevel()                                               | Set graph level                                |
| clear()/show()                                           | Clear or transmit frame                        |
| setBrightness()                                          | Set one LED PWM                                |
| setBrightnessPercent()                                   | Set one LED percentage                         |
| setLED()                                                 | Set one LED 0–100%                             |
| getBrightness()/getBrightnessPercent()                   | Read one LED brightness                        |
| setGlobalBrightness()                                    | Set overall brightness                         |
| flashLED()                                               | Flash one LED; optional timing/peak brightness |
| isFlashing()                                             | Read flash state                               |
| setGraduated()/setGraduationMin()                        | Graduated brightness control                   |
| setGreenToRed()                                          | Set logical direction                          |
| setTransition()/setTransitionSpeed()/setTransitionTime() | Transition control                             |
| isTransitioning()                                        | Read transition state                          |
| setEffectSpeed()                                         | Set effect speed                               |
| startEffect()/stopEffect()                               | Control effects                                |
| isEffectRunning()                                        | Read effect state                              |
| setPulseRange()                                          | Set Pulse/Breathe limits                       |
| update()                                                 | Service non-blocking animations                |

## 16. Version History

V1.6 — Moving Dot effect.

V1.7 — Knight Rider effect.

V1.8 — Pulse/Breathe effect.

V1.9 — Global brightness and process-indicator flashing.

V2.2.x — Refined individual LED flashing and manual update operation.

V2.3.0 — Added setLED(index, percent) for individual LED on/off and percentage brightness.